Lecture plan · for the lecturer

Public-key cryptography

Cyber security · University · Lecture 2

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Run Diffie–Hellman.
- Explain RSA encryption and signatures.
- Expose the man in the middle.

Activity: Alice, Bob and Eve

1. Warm-up (5 min): كيف يمكن ذلك؟
2. Mix (10 min): Show the paint-mixing analogy and why unmixing is hard.
3. Exchange (20 min): Groups of three run DH with p = 23; Eve records everything she hears.
4. Attack (20 min): Replace Eve with an active attacker, then fix it with signatures.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Why couldn’t Eve find the key?
- What did signatures add?
- Which misconception did you hear today?

Exit ticket

- Say what encrypting to Bob’s public key does and does not guarantee.

Try it in your next lecture
Put students in threes as Alice, Bob and Eve; they run Diffie–Hellman with p = 23 on paper while Eve tries to find the key.

Seminar sheet · for students

Public-key cryptography

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Name _____ Class _____ Date _____

1 **Plain Diffie–Hellman has no authentication. What can a network attacker do?**

◯ Nothing, the key stays secret ◯ Run an exchange with each side and relay messages

◯ Factor p

2 **Run Diffie–Hellman with p = 23, g = 5.**

Alice picks a = 3, so A = 5^3 mod 23 = _____ . Bob sends B = 10, so the key is 10^3 mod 23 = _____ .

Word bank: 10 · 15 · 8 · 11

3 **Which statements are true? Choose all that apply.**

☐ A signature made with Alice's private key can be verified by anyone

☐ Encrypting to Bob's public key proves Alice sent it ☐ Textbook RSA without padding is insecure

☐ TLS authenticates the key exchange with certificates

4 **A longer RSA key fixes a man-in-the-middle attack on an unauthenticated exchange.**

◯ True ◯ False

5 **Match each goal to the operation that provides it.**

Only Bob can read it	• • • •	A certificate
Anyone can check Alice wrote it	• • • •	Diffie–Hellman
Agree a key in public	• • • •	Sign with Alice's private key
Bind a public key to a name	• • • •	Encrypt with Bob's public key

✎ **How much number theory do your students need before they meet public-key cryptography? What could they do by hand first?**